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CLOUD COMPUTING

Session 1

Arun Vadekkedhil

Session Agenda

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AGENDA

Cloud Computing – An Overview



“ Let's explore the cloud fundamentals and build a strong foundation together!



01 Cloud Computing – An Overview

- Introductions – Tutor, Course & Students
- Paradigms & Distributed Computing
- Origins & Motivation for Cloud
- What is Cloud Computing
- Is Cloud Computing for me?



02 Types of Clouds & Service Deployment

- 3-4-5 Rule of Cloud Computing
- Cloud Infrastructure & Deployment



03 Wrap Up

- Perceived benefits of Cloud ecosystem
- Challenges to Overcome
- Advantages & Disadvantages
- Commercial offering of Cloud services
- Reality Check



Goal: Understand the cloud ecosystem, its value, and how it powers the modern digital world.



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Introductions

*Getting to know each other,
building a strong learning community.*



Introductions

Let's learn, explore and build the cloud future together!



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About Me



Name

Prof. Arun Vadekkedhil



Profession

Solutions Architect
(GenAI, RPA & Cloud)



Interests

Tutoring, writing



Contact

9881300394



Graduated from BITS Pilani in 1995, have over **24+ years** of IT experience, Tutor for the past **8 years**.



About the Course



Units: 5

Objective:

No	Course Objective
CO1	Students will learn the fundamental ideas behind Cloud Computing, the evolution of the paradigm, its applicability; benefits, as well as current and future challenges.
CO2	Students will learn the basic ideas and principles in data centre design and management.
CO3	Students will learn about cloud components and technologies and relevant distributed file systems.
CO4	Students will learn a variety of programming models and develop working experience.

Evaluation Components



02

Quizzes



02

Written Exams



01

Assignment

Ready to explore the cloud? **Let's get started!**

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Cloud – An Introduction

Making the jump. New terms, new perspectives.



What is Cloud Computing

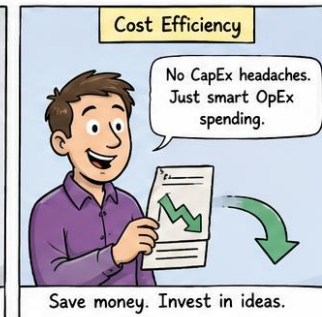
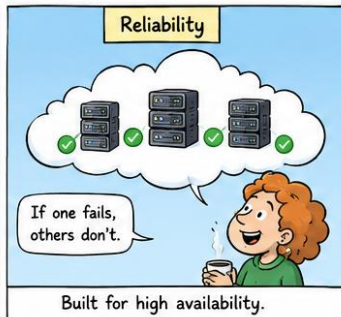
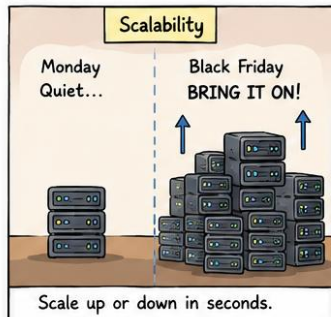
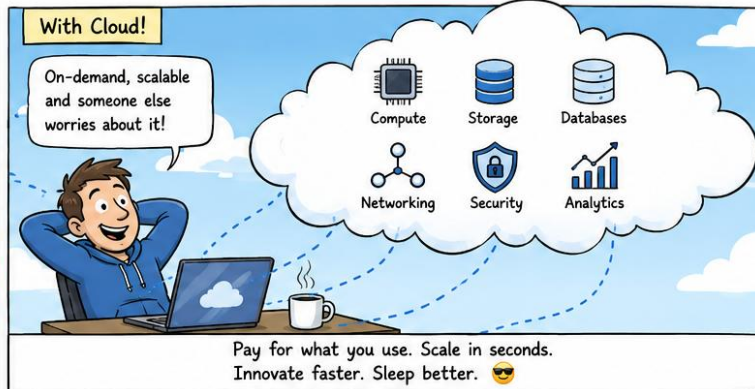
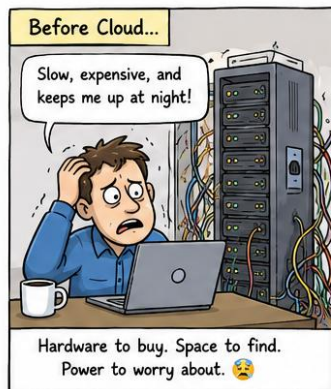
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Cloud Computing Explained

It's not magic... just someone else's computer.



Cloud Computing = Agility + Innovation + Security + Reliability + Cost Efficiency

Focus on your business. We handle the cloud. 😎



Who is a Cloud Engineer

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Cloud Engineer Skills at a Glance

Building scalable, secure and intelligent cloud solutions



Core Mindset: Security First



Think Automation



Embrace Continuous Learning



Build for Scale & Reliability

Who is a Cloud Engineer

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 Traditional Cloud Engineer	Focus Area	 Modern AI-Era Cloud Engineer
 Deploy virtual machines	 Compute	Deploy AI platforms and GPU clusters 
 Configure networking	 Networking & Data	Build secure AI networking and data pipelines 
 Manage servers	 Platforms	Manage Kubernetes and serverless platforms 
 Write deployment scripts	 CI/CD	Build AI-powered CI/CD pipelines 
 Monitor infrastructure	 Monitoring	Monitor AI models and infrastructure 
 Provision storage	 Data & Storage	Design data lakes and vector databases 
 Focus on automation	 Automation	Use AI agents for automation 
 Manage cloud resources	 Cloud Architecture	Build AI-ready cloud architectures 

Facts about Cloud

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CLOUD FACTS 2025

Key insights shaping the modern cloud ecosystem



of enterprises use cloud services in some form.

Source: Flexera 2024



Global cloud market expected to reach \$1.03 trillion in 2025.

Source: Gartner 2024



of organizations say cloud security is a top priority.

Source: Microsoft 2024



of data will be stored in the cloud by 2025 (1 ZB = 1 trillion GB).

Source: IDC 2024



of workloads will be in the cloud by 2025.

Source: Gartner 2024

THE MODERN CLOUD ECOSYSTEM



Hybrid & Multi-Cloud

Organizations use a mix of public, private and edge clouds for flexibility and control.



AI Everywhere

Cloud powers AI/ML workloads, copilots, and intelligent applications at scale.



Cloud-Native

Containers, microservices and serverless drive agility and faster innovation.



Security First

Zero Trust, identity security and compliance are built into the cloud.



Data-Centric

Data lakes, real-time analytics and vector databases unlock insights.



Edge Expansion

Edge computing brings cloud capabilities closer to users and devices.



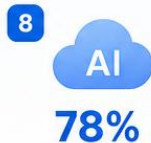
of organizations are optimizing cloud spend through FinOps.

Source: Flexera 2024



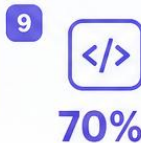
increase in public cloud spend in 2024 compared to 2023.

Source: Canalys 2024



of organizations use cloud for AI/ML or plan to within 18 months.

Source: Deloitte 2024



of applications built using cloud-native technologies.

Source: CNCF 2024



of enterprises have a multi-cloud strategy.

Source: Flexera 2024

WHY IT MATTERS



Cloud enables businesses to innovate faster, operate securely, and scale without limits—driving growth in a digital-first world.

WHAT'S DRIVING CLOUD ADOPTION?



Digital Transformation



Cost Optimization



Business Agility



Resilience & Business Continuity



Data & AI Innovation



Sustainability Goals



The cloud is not just infrastructure—it's the foundation for innovation, resilience and sustainable growth.

Sources: Flexera 2024, Gartner 2024, IDC 2024, Microsoft 2024, Canalys 2024, Deloitte 2024, CNCF 2024

Why is it Called Cloud

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THE CLOUD COMPUTING UNIVERSE

Powering the World, One Click at a Time

The Cloud Symbol

Eponymously named after the cloud symbol used in architecture documents.

By now, you must be aware that this has no relation to its namesake from the meteorology department, **but it simply means that we are using the internet to store data on remote servers**, rather than storing them locally on our hard disks.

Many types of Cloud Computing Applications Exist.

Depending on the need of your IT the type of cloud solution may vary.

Cloud Computing is up to **40 times more cost-effective** for an SMB compared to running its own IT system or department.

ON-PREMISES IT

High upfront cost, hardware, maintenance, power, cooling, IT staff & infrastructure.



UP TO
40x
MORE
COST-EFFECTIVE



CLOUD COMPUTING

Lower cost, pay-as-you-go, no maintenance, automatic updates & global reach.

TYPES OF CLOUD COMPUTING APPLICATIONS



PUBLIC CLOUD

Services delivered over the internet and shared by multiple customers.



PRIVATE CLOUD

Cloud resources dedicated to a single organization.



HYBRID CLOUD

Combination of public and private clouds for greater flexibility.



COMMUNITY CLOUD

Shared by several organizations with common concerns.



MULTI-CLOUD

Using multiple cloud services from different providers.



The Cloud isn't just a destination. It's a smarter way to work, connect and grow.

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Cloud – Origins & Motivation

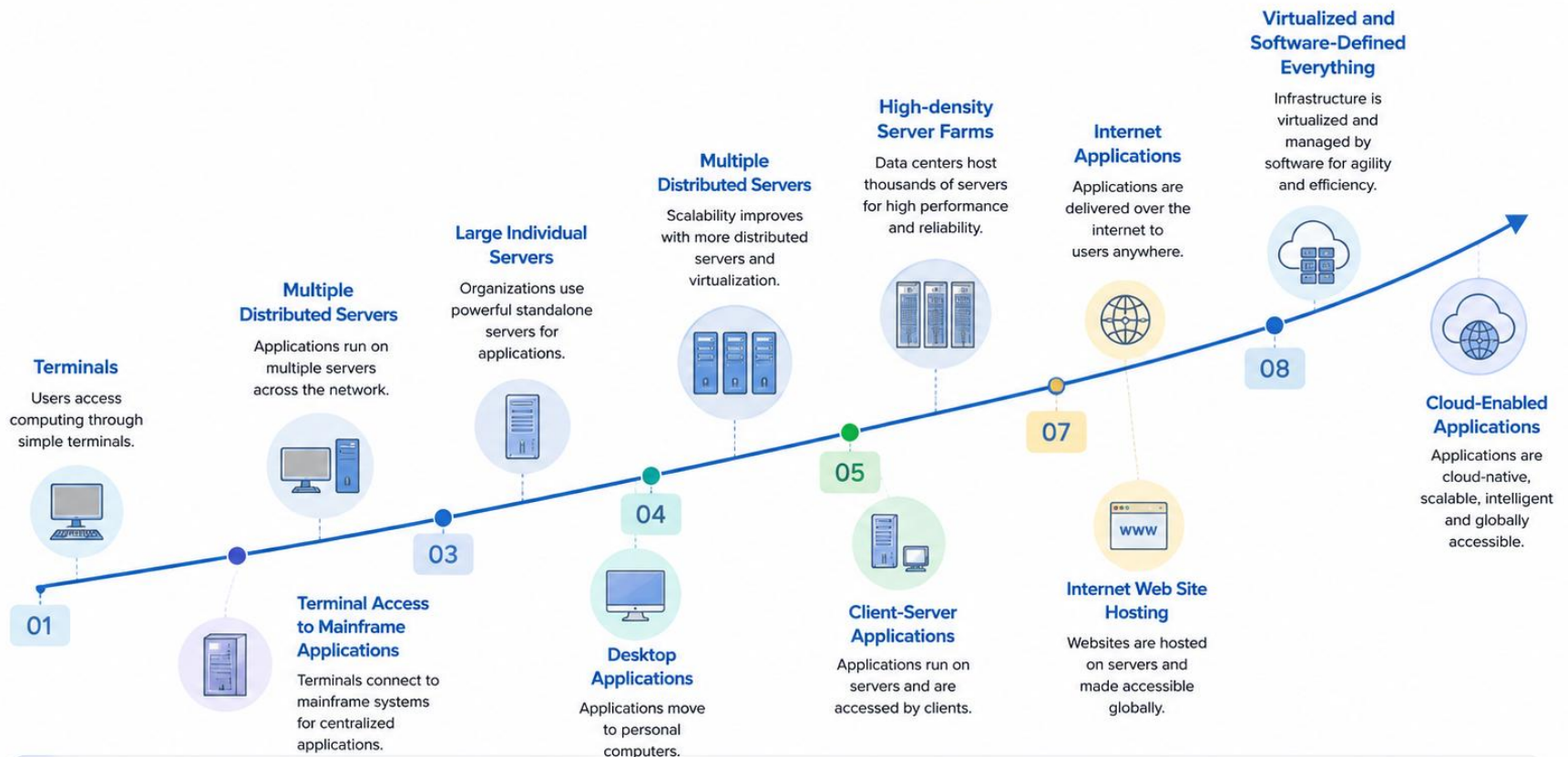
In the beginning, space was expensive and sparingly used.



- Traditional computational models are now passé. The era of machines being hosted in-house or as we call in cloud on premise is seeing a marked shift.
- Amazon started the concept by renting spare computing power from their retail business.
- Web and mobile technologies have resulted in information explosion. This means traditional computing power is not enough to process data.
- To solve this challenge, we need to use large scale distributed networks. These distributed networks evolved to the Cloud technology.

THE EVOLUTION OF COMPUTING

From Terminals to Cloud-Enabled Applications



From simple terminals to intelligent cloud applications – computing has evolved to empower businesses and people everywhere.



Paradigms

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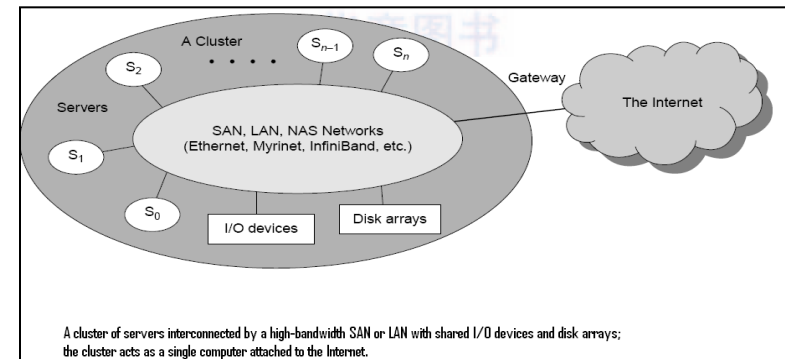
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A *paradigm* is a standard, perspective, or set of ideas. A *paradigm* is a way of looking at something.

Types of Distributed Computing

- **Parallel Computing** : Parallel computing is a form of computation in which many calculations are carried out simultaneously, operating on the principle that large problems can often be divided into smaller ones, which are then solved concurrently ("in parallel")
- **Cluster** : A cluster is a group of loosely coupled computers that work together closely, so that in some respects they can be regarded as a single computer.
- **Grids** : Grid computing is the most distributed form of parallel computing. It makes use of computers communicating over the Internet to work on a given problem.



Paradigms



Cloud is the convergence of several traditional computing technologies

Web Technologies

Web Services

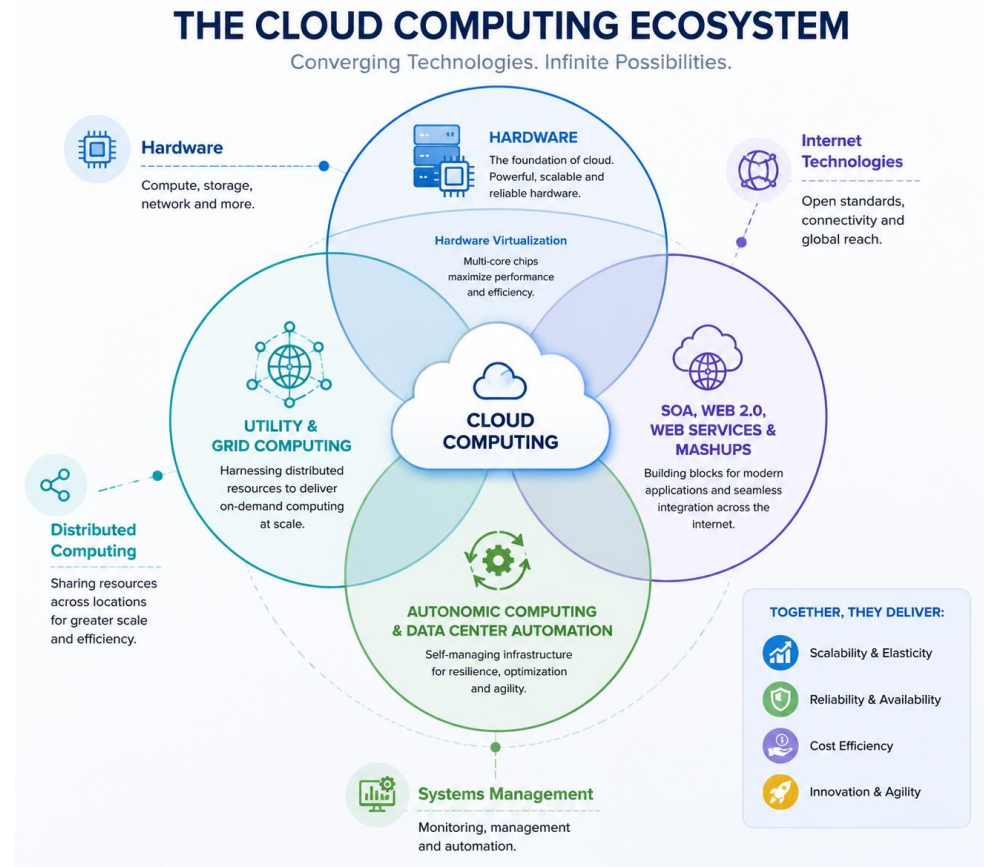
SOA (Service Oriented Architectures)

Distributed Computing

Grids

Clusters

We are now experiencing computing as a service provided by professional organizations, which is served through the medium of high speed internet.

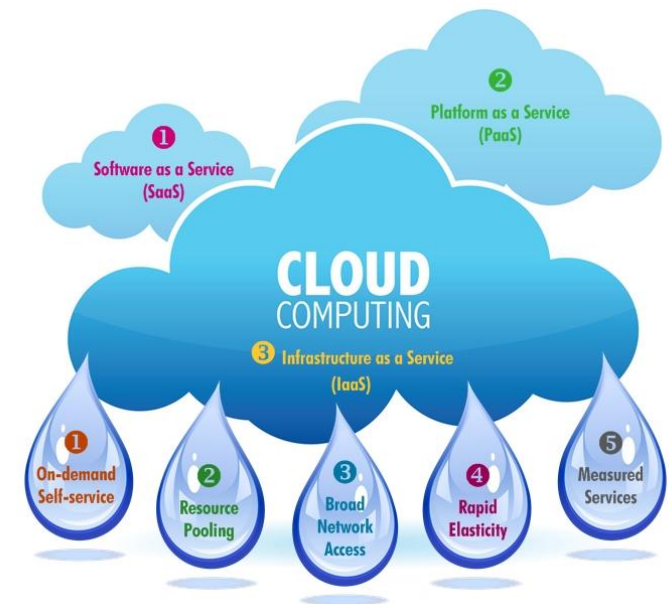


How do we define Cloud ?



To summarize, cloud computing is the result of the mash-up of several existing disparate technologies, which were progressively modified to suit contemporary computing requirements.

- The applications and services that run on a distributed network using virtualized resources and accessed by common Internet protocols and networking standards comes under Cloud computing. Cloud computing converts the technology, services, and applications that are similar to those on the Internet into a self-service utility. Communicate and coordinate actions by passing messages.
- Cloud computing is based on the concept of pooling physical resources and presenting them as a virtual resource. This computing model supports a new way of provisioning resources, staging applications and for using applications.
- It's bringing computing on an internet scale.



Terms to Remember



Cloud computing solely exists because of Virtualization technology & Abstraction

Abstraction :

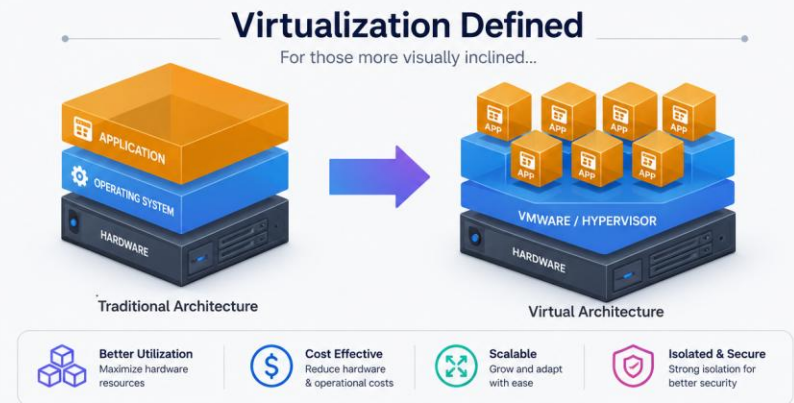
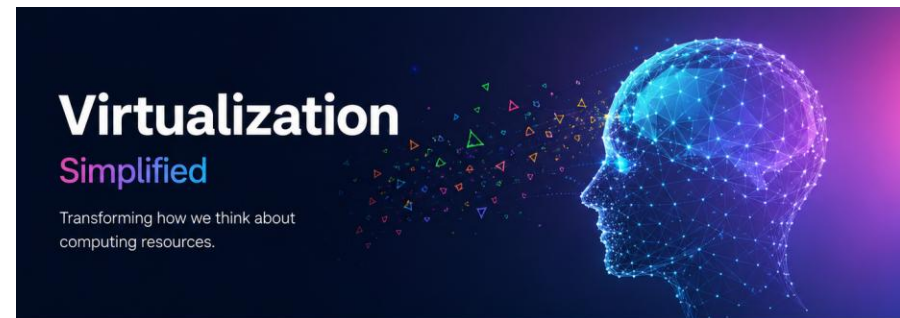
The details of system implementation is hidden from users and developers.

Applications run on unspecified physical systems with unknown locations for data, with outsourced system administration of systems.

Virtualization:

The resourced are pooled and shared among the users giving them the illusion that they are the sole owner of the resource.

Also resources scales up/down in really short time and without human intervention, charged on metered basis, with multi-tenancy support.



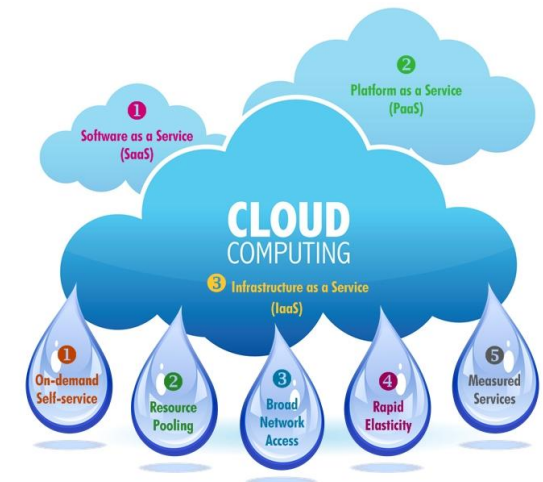


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Cloud – NIST & the AAS es

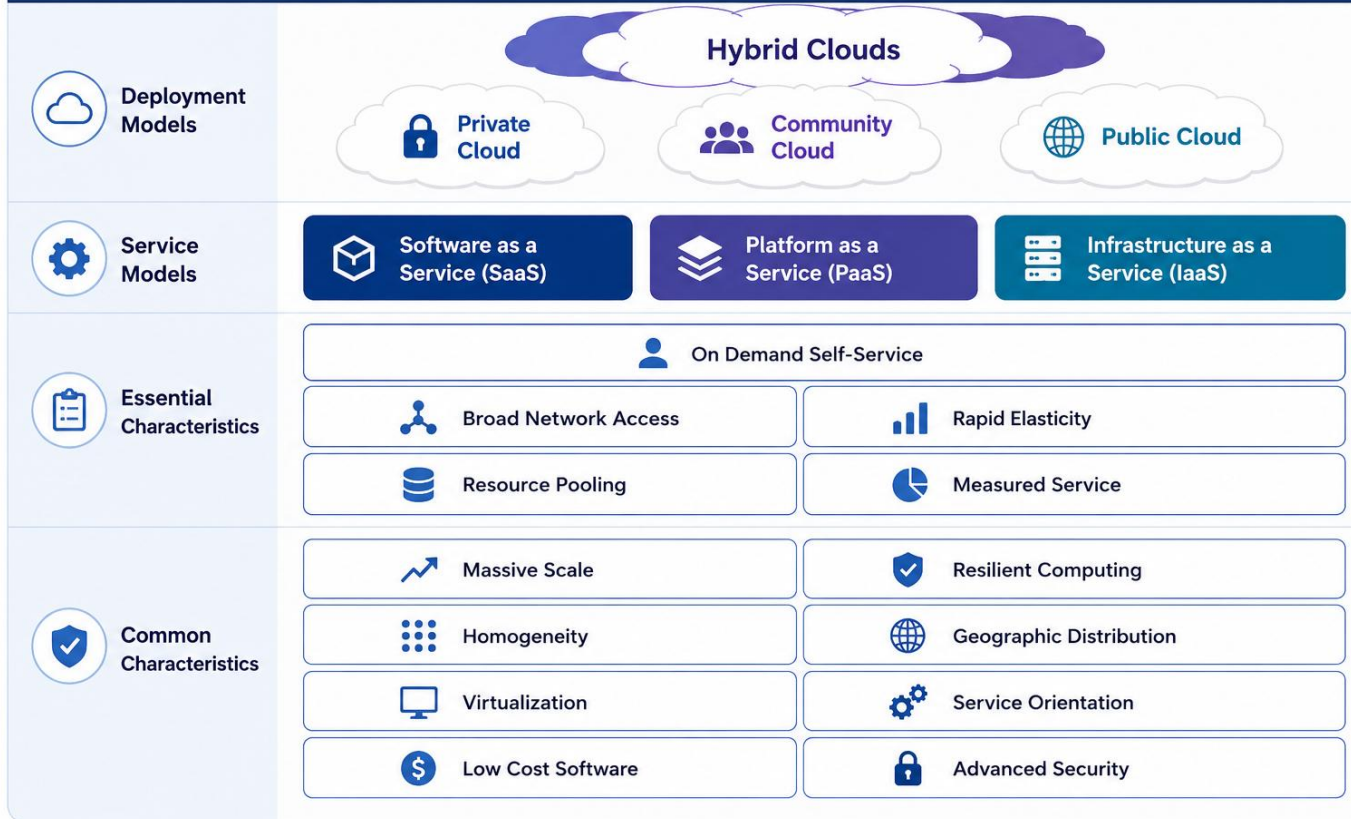
Getting serious now, some times we cant skip the rote process. Ready to challenge ye faithful ole grey cells?



NIST 3-4-5 Rule for Cloud



The NIST Cloud Definition Framework



- **3** cloud service models or service types for any cloud platform
- **4** Deployment models
- **5** Essential characteristics of cloud computing infrastructure

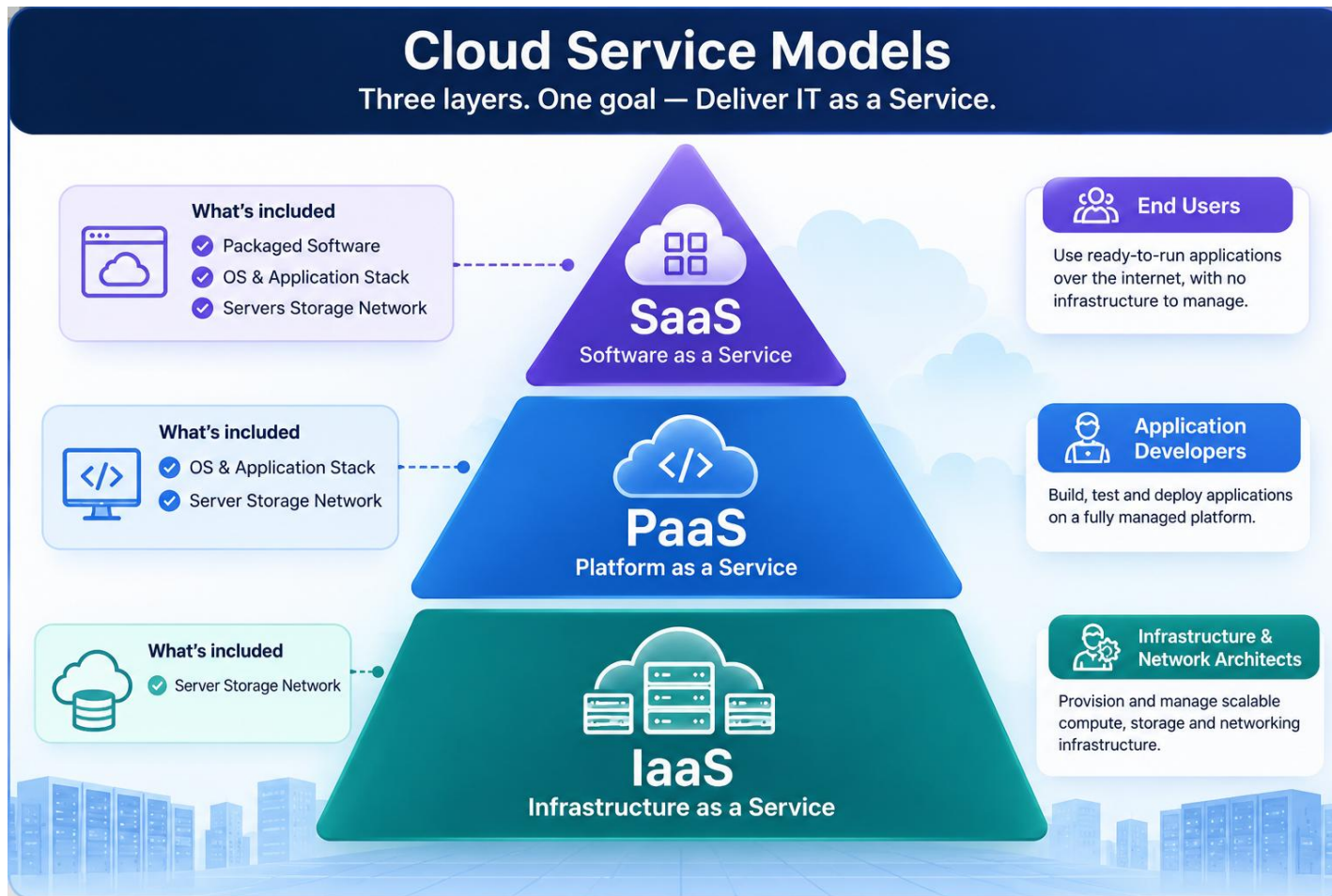
The applications and services that run on a distributed network using virtualized resources and accessed by common Internet protocols and networking standards comes under Cloud computing.

Cloud Service Models (AAS es)

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- There are 3 service models
- **Infrastructure** as a Service
- **Platform** as a Service
- **Software** as a Service

Infrastructure as a Service

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Why IaaS



Capability

IaaS provides the following

- Servers- compute, machines
- Storage
- Network
- Operating system



IaaS delivers the foundational building blocks to run IT workloads in the cloud.



IAAS

Infrastructure as a Service

Characteristics



Resources are distributed as a service

- Allows dynamic scaling (1...10...100....)
- Has a variable costs-
- Generally includes multiple-users on a single piece of hardware. (**multi-tenancy**)



Scale up or down on demand and pay only for what you use.



Models

IaaS can be obtained as

- 1 Public or
- 2 Private infrastructure or
- 3 combination of both



Choose the right model for your business needs and compliance requirements.

Benefit



The user instead of **purchasing**-servers, software, **data center space** or network equipment, **rent** those resources as a fully **outsourced** service on-demand model.



Reduce CapEx. Increase agility. Focus on innovation, not infrastructure.

Enabler : Virtualization Technology



Manageability and Interoperability



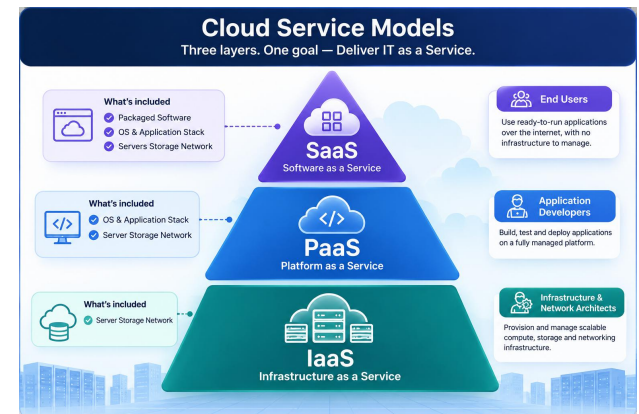
Availability and Reliability



Scalability and Elasticity

Infrastructure as a Service - Definition

- The capability through which the consumer can provision processing, storage, networks, and other required computing resources.
- The consumer can deploy and run software, which can include operating systems and applications.
- The consumer does not manage or control the underlying cloud infrastructure but has control over operating systems, storage, deployed applications, and possibly limited control of select networking components (e.g., host firewalls).
- Offering virtualized resources (computation, storage, and communication) on demand is known as Infrastructure as a Service (IaaS).
- Infrastructure services are considered to be the bottom layer of cloud computing systems .
- Ex : Amazon EC2 : Elastic Compute Cloud, Eucalyptus, GoGrid, Rackspace Cloud



Infrastructure as a Service - Applicability

Where IaaS Helps

-  Need **computing power on demand**
-  Employees work **on the go**
-  Business experiences **sudden traffic spikes**
-  Development & Testing
-  Growing storage requirements
-  Backup & Disaster Recovery
-  Hosting web or enterprise applications
-  Expanding into new locations
-  High capital expenditure on IT
-  AI/ML & High Performance Computing

What You Can Do with IaaS

- Provision virtual machines in minutes instead of weeks
- Access applications securely from anywhere
- Scale infrastructure up or down automatically
- Create temporary environments and delete them when finished
- Expand storage instantly without buying new hardware
- Replicate workloads across regions for business continuity
- Deploy highly available applications with global reach
- Launch infrastructure without building a new data center
- Convert CapEx into predictable Pay-as-you-go OpEx
- Rent GPU and high-performance compute only when required

Platform as a Service

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Why PaaS



Capability

PaaS provides the following

- Tools to build applications
- Scripting Environment
- Database Platform



Everything you need to build, run and manage applications.



Models

PaaS can be obtained as

- 1 Public or
- 2 Private infrastructure or
- 3 combination of both



Flexible deployment options to match your business and compliance needs.



Characteristics

Collaborative platform for application development using workflows.

Platform which allows creation of proprietary data or application



Build better applications faster with collaboration and automation.



Benefit

Development tools served up on a Platter a-la carte

No need to worry about upgrading to newer platforms or worry about license costs



Lower costs, faster delivery, focus on innovation.

Enabler : Runtime Environment Design



Fault Tolerant Design



Containerization



Avoiding DLL Hell



Secure



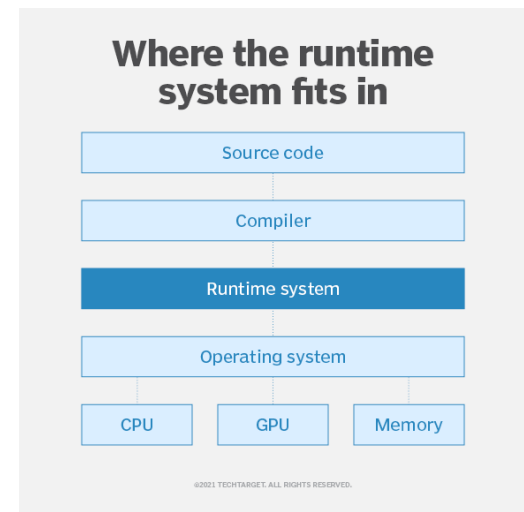
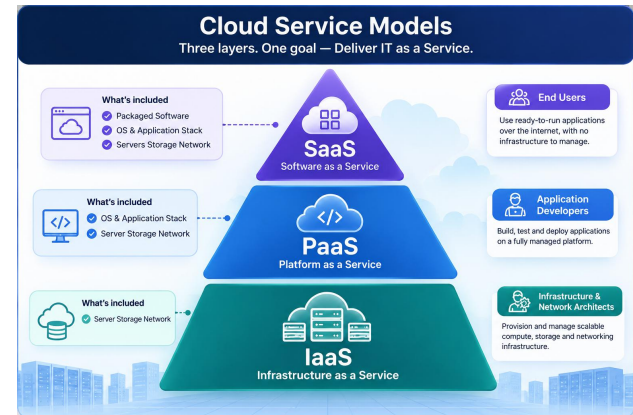
In Summary

PaaS empowers developers to build, deploy, and manage applications efficiently without worrying about the underlying infrastructure.

Platform as a Service - Definition



- The capability provided to the consumer is to **deploy** onto the **cloud infrastructure**, **consumer-created** or acquired applications created using programming languages and **tools supported by the provider**.
- The consumer **does not manage** or control the underlying **cloud infrastructure** including **network**, **servers**, **operating systems**, or **storage**, but has control over the **deployed applications** and possibly application **hosting** environment **configurations**.
- A **PaaS platform** offers an environment on which developers **create** and **deploy applications** and do not **necessarily** need to **know** how many **processors** or how much **memory** that **applications** will be **using**.
- In addition, multiple programming models and **specialized services** (e.g., data access, authentication, and payments) are offered as building blocks to new applications.
- Google AppEngine**, **Azure**, **Force.com** an example of Platform as a Service



Platform as a Service - Applicability

Section	Description	Key Benefits
✂ Platform Capabilities	• Allows developers to frequently change or upgrade operating system features. • Provides development tools, runtime, middleware, and managed services.	Faster development without managing infrastructure
👥 Collaboration	• Enables development teams to collaborate on projects. • Shared development environments for multiple team members. • Tools for billing and subscription management.	Improved teamwork and productivity
🔒 Security & Reliability	• Built-in security mechanisms. • Data security, backup, and recovery are managed by the platform provider.	Secure and resilient application development
🔄 Adaptability	• Platform features can be modified or updated as business requirements evolve.	Rapid response to changing business needs
⚙ Flexibility	• Customers control the development tools installed within their platform. • Developers can customize the environment to suit project requirements.	Personalized development experience
🚀 Application Services	• Services to develop, test, deploy, host, and maintain applications in a single integrated development environment. • Web-based UI creation tools simplify application development, testing, and deployment.	Accelerates the complete application lifecycle
👥 Multi-Tenancy	• Multiple concurrent users can utilize the same development platform securely and efficiently.	Better resource utilization and lower costs

Software as a Service

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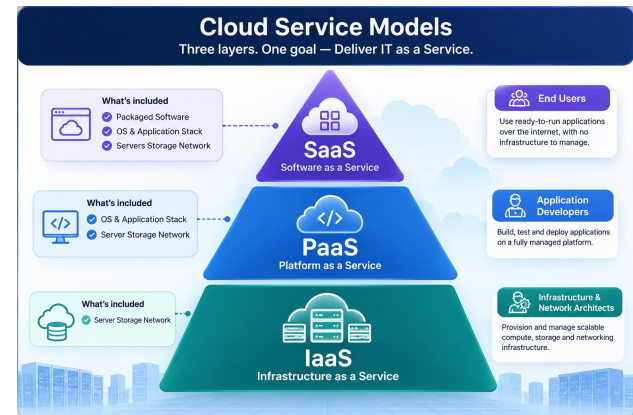
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Software as a Service - Definition

- The capability provided to the consumer is to use the provider's applications running on a cloud infrastructure.
- The applications are accessible from various client devices through a thin client interface such as a web browser (e.g., web-based email).
- The consumer does not manage or control the underlying cloud infrastructure including network, servers, operating systems, storage, or even individual application capabilities, with the possible exception of limited user-specific application configuration settings.
- This model of delivering applications, known as Software as a Service (SaaS), alleviates the burden of software maintenance for customers and simplifies development and testing for providers.
- Salesforce.com, SaaS model, offers business productivity applications (CRM) that reside completely on their servers, allowing customers to customize and access applications on demand.



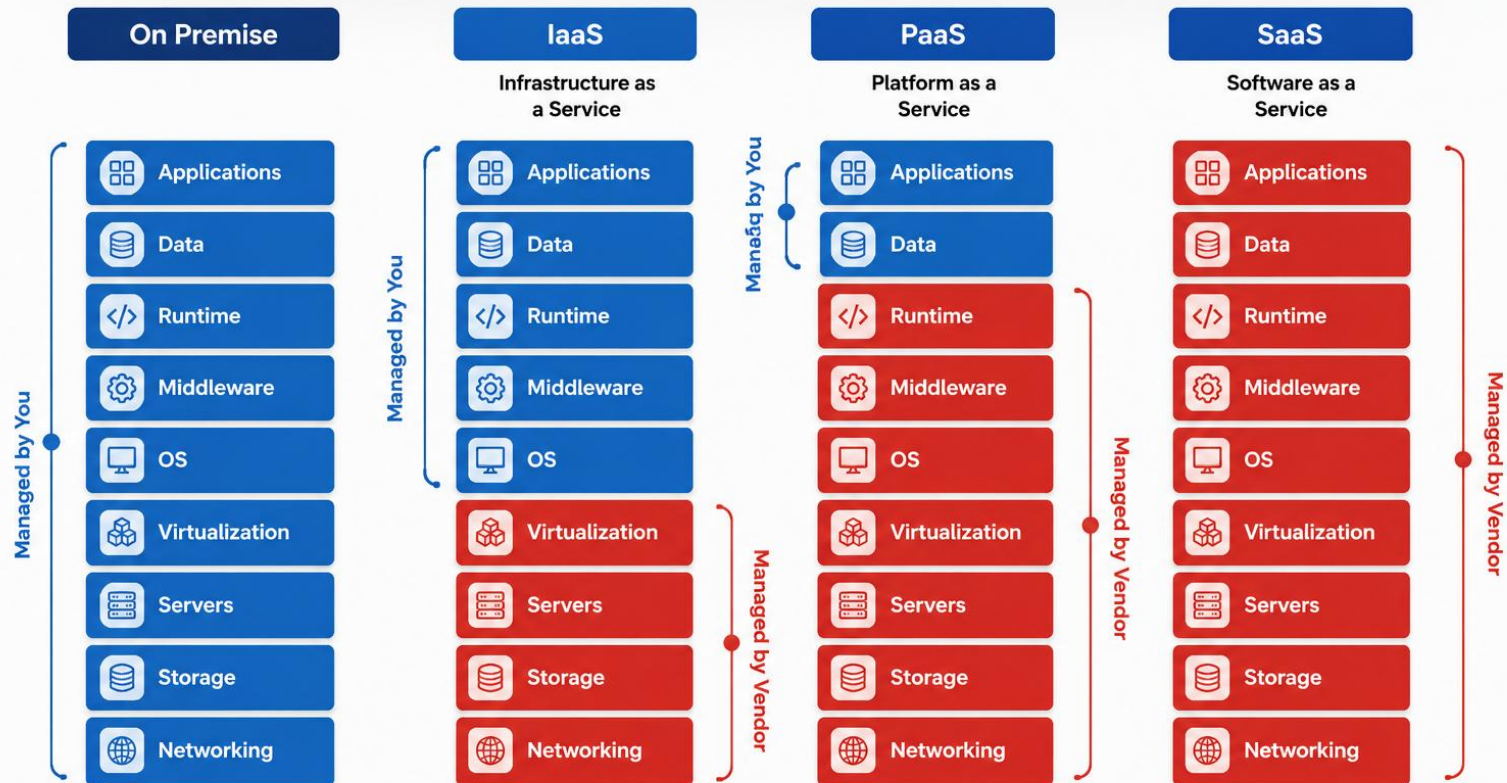
Which AAS ?

WHICH CLOUD COMPUTING MODEL SHOULD YOU CHOOSE?

IAAS	PAAS	SAAS
Cover infrastructure maintenance and support	Focus on app development instead of infrastructure management	Create solutions with standardized core functionality
Save money and time vs purchasing hardware	Streamline workflows when multiple developers are working on the same project	Develop ecommerce software rapidly, without spending time on server or software issues
Achieve flexibility and scalability when experiencing rapid growth	Rapidly launch an application, reducing costs and time spent on hardware and middleware management	Work on short-term projects and applications that need both web and mobile access

www.apriorit.com

Who Manages the AAS es?



Managed by You

You are responsible for managing and securing these layers.

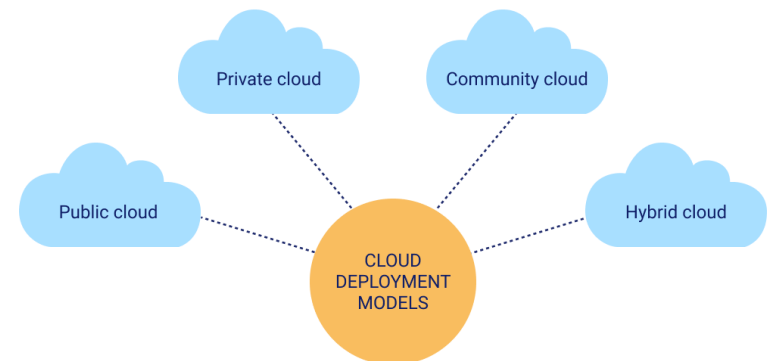


Managed by Vendor

The cloud provider manages and secures these layers.



Cloud Deployment Models

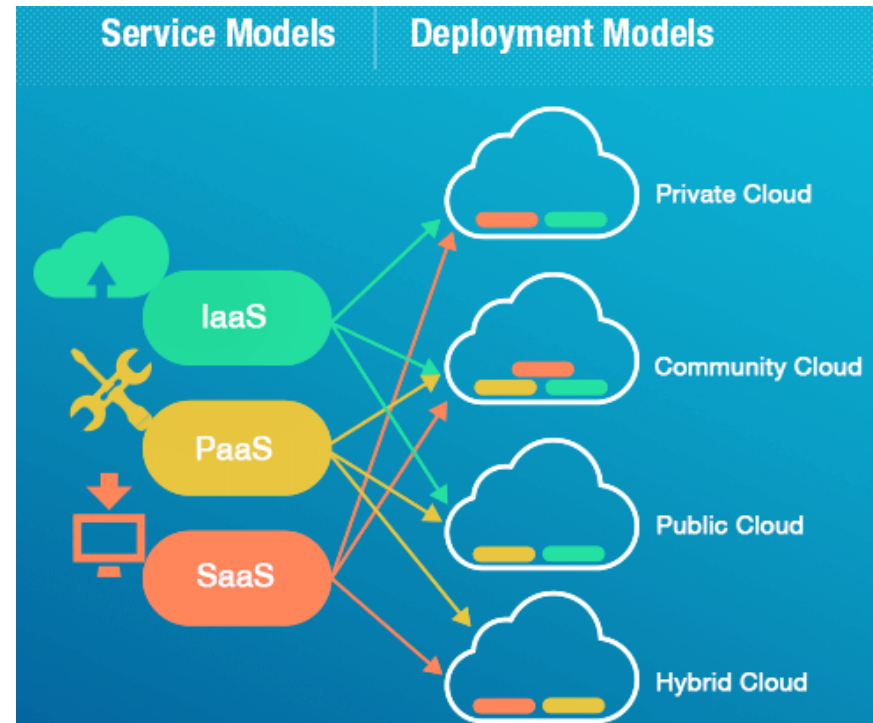


Deployment Models in Cloud

There are four primary cloud deployment models :

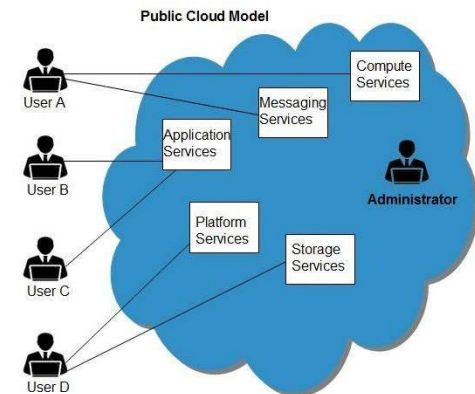
- Public Cloud
- Private Cloud
- Community Cloud
- Hybrid Cloud

Each can exhibit the previously discussed characteristics; their differences lie primarily in the scope and access of published cloud services, as they are made available to service consumers.



Pubic Cloud

- **Public cloud** is a cloud infrastructure **owned by a cloud service provider** that provides cloud services to the public for **commercial purposes**.
- Cloud infrastructure available for public consumption on a **pay per use** basis.
- Examples of public clouds include **Amazon Elastic Compute Cloud (EC2)**, IBM's Blue Cloud, Sun Cloud, Google AppEngine and Windows Azure Services Platform.
- **Characteristics**
 - ✓ Homogeneous infrastructure
 - ✓ Common policies , Shared resources and multi-tenant
 - ✓ Leased or rented infrastructure, Economies of scale



Private Cloud

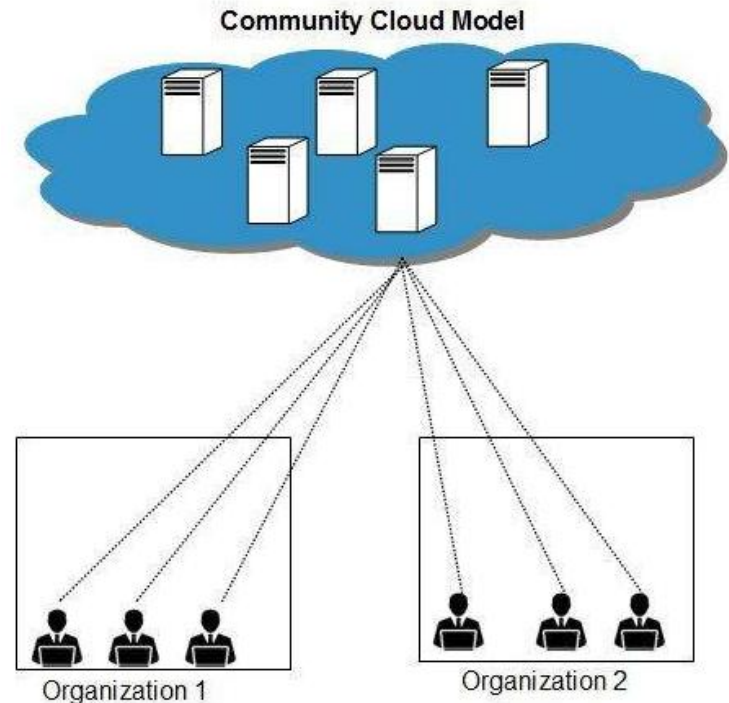
- A Private Cloud is a **cloud infrastructure** that is exclusively dedicated to a single organization.
- The infrastructure can be owned, managed, and operated by the organization itself or by a trusted third-party provider, and it may be deployed on-premises, in a colocation facility, or in a hosted data center.
- Unlike public clouds, a private cloud provides dedicated computing resources that are not shared with other organizations, offering greater control, security, customization, and compliance.
- Private clouds are commonly adopted by organizations that handle sensitive data, have strict regulatory requirements, or require customized infrastructure and security policies..

Hybrid Cloud

- A hybrid cloud combines two or more distinct cloud environments (private cloud, public cloud, or on-premises infrastructure) that remain separate but are connected to work as a unified environment.
- Organizations can run workloads where they make the most sense.
- For example, sensitive customer data may remain in a private cloud or on-premises data center, while customer-facing web applications and AI workloads run in a public cloud.
- Resources can be dynamically extended to the public cloud during periods of high demand (known as cloud bursting), helping organizations scale without investing in additional hardware.
- Each environment retains its own identity, but they are integrated through secure networking, unified management, and orchestration.
- Hybrid clouds support application portability, data integration, disaster recovery, and business continuity, allowing applications and workloads to move between environments with minimal disruption.

Community Cloud

- A Community Cloud is a cloud infrastructure that is shared by multiple organizations with common business, security, compliance, or operational requirements.
- The participating organizations collaborate by sharing cloud resources while maintaining agreed-upon governance and security standards.
- Unlike a Public Cloud, access is restricted only to the participating organizations.
- Unlike a Private Cloud, the infrastructure is shared among multiple organizations rather than being dedicated to a single enterprise.
- The cloud infrastructure may be owned, managed, and operated jointly by the participating organizations or by a trusted third-party cloud provider, and it may be deployed on-premises or in a hosted data center.



Deployment Models - Advantages & Disadvantages

CLOUD DEPLOYMENT MODELS: PROS AND CONS



PUBLIC CLOUD

Services over the public internet, owned and operated by a CSP

ADVANTAGES

- ✓ No upfront capital expenditure, pure pay-as-you-go
- ✓ Elastic, near-instant scalability
- ✓ No hardware/facility maintenance burden on the customer
- ✓ Global reach via multiple provider regions
- ✓ Fast provisioning, minimal lead time

DISADVANTAGES

- ✗ Vendor lock-in through proprietary APIs and managed services
- ✗ Egress fees make moving data back out expensive
- ✗ Cost can become unpredictable and, at steady-state scale, can exceed private infrastructure TCO
- ✗ Limited control over underlying hardware, patch schedules, physical placement
- ✗ Compliance/data-residency constraints in regulated sectors (finance, health, government)
- ✗ Multi-tenancy risk exists, but "public cloud = less secure" is misleading—most breaches trace to customer-side misconfiguration



PRIVATE CLOUD

Dedicated infrastructure for a single organization

ADVANTAGES

- ✓ Full control over infrastructure, stack, and customization
- ✓ Data stays on-premises or in dedicated infra — simplifies compliance narrative
- ✓ No "noisy neighbor" multi-tenancy interference
- ✓ Security policy fully under organizational control

DISADVANTAGES

- ✗ High capital expenditure; must provision for peak load, not average
- ✗ Limited elasticity compared to public cloud; scaling means buying hardware
- ✗ Requires in-house infrastructure and ops expertise
- ✗ Slower to provision new capacity than public cloud
- ✗ Idle-capacity risk — you pay for peak-sized infrastructure even during low-demand periods



HYBRID CLOUD

Combination of private cloud and public cloud with integration

ADVANTAGES

- ✓ Combines public elasticity with private control for sensitive workloads
- ✓ Supports cloud bursting — scale into public cloud during demand spikes, release after
- ✓ Sensitive data can stay private while customer-facing apps run in public cloud
- ✓ Enables disaster recovery and business continuity flexibility

DISADVANTAGES

- ✗ Networking complexity from integrating two distinct environments
- ✗ Requires unified management/orchestration tooling across environments
- ✗ Maintaining consistent security policy across two environments is harder than in either alone
- ✗ Still dependent on internal IT infrastructure reliability — hybrid doesn't remove that dependency, it adds a second one



COMMUNITY CLOUD

Shared by organizations with common compliance, security or mission needs

ADVANTAGES

- ✓ Shared infrastructure among organizations with common compliance/security/mission needs, so no single org bears full cost
- ✓ Access restricted to the consortium, not the general public — narrower attack surface than public cloud
- ✓ Sits between private and public on the security/control spectrum — tighter than public multi-tenancy, without the full cost of dedicated private infrastructure
- ✓ Enables collaboration that pure private cloud can't — e.g., joint data access, shared research infrastructure

DISADVANTAGES

- ✗ Governance is genuinely hard — allocating responsibility for security, cost, and operations across multiple independent organizations is a recurring failure point
- ✗ Shared physical/logical infrastructure means one participant's data sits alongside others' — requires careful tenant isolation, and a breach or dispute involving one member can implicate the group
- ✗ Slower and more expensive to stand up than either alternative — negotiating shared SLAs, joint audit trails, and federated identity takes time
- ✗ Scalability is capped by the consortium's combined infrastructure, not elastic like public cloud



KEY TAKEAWAY

There is no one-size-fits-all cloud. Choose the model that best fits your workload, compliance needs, control requirements, and cost expectations. Many organizations use a mix of models.

✓ Strength / Advantage

✗ Challenge / Disadvantage

Quick Comparison

Model	Think of it as...	Primary Strength	Primary Trade-off
 Public	Rent an apartment	Scale & speed	Less infrastructure control
 Private	Own your house	Control & customization	Higher cost & responsibility
 Hybrid	Own a house + rent when needed	Flexibility	Integration complexity
 Community	Shared facility for a group	Common requirements & collaboration	Shared governance

Multi Cloud

The New Default, Not the Exception



What Is Multi-Cloud?

The deliberate use of two or more public cloud providers (e.g., AWS, Azure, GCP) to run different workloads — rather than standardizing on one vendor for everything.

90%+

of enterprises now run a multi-cloud strategy

Source: Flexera 2024 State of the Cloud Report



WHY ORGANIZATIONS USE IT



Avoid Lock-in

Preserves negotiation leverage and exit options with any single cloud vendor.



Best-of-Breed

Pick the strongest services you need — e.g., AI/ML on one cloud, core apps on another.



Resilience

Architect with built-in regional or provider diversity to reduce downtime and eliminate single points of failure.



Data Residency

Keep data within required countries or regions to meet regulatory and industry needs.



Most enterprises choose multi-cloud as a strategic enabler — it also arrives through M&A, modern IT, and business units piloting different platforms. Either way, managing it deliberately beats inheriting it by accident.

Multi Cloud

Advantages vs. Disadvantages

ADVANTAGES



Negotiating Leverage

No single vendor controls pricing or roadmap; contracts can be renegotiated with real alternatives on the table.



Best-of-Breed Services

Teams use each provider's strongest offering instead of settling for one vendor's weakest link.



Resilience & Redundancy

Workloads can fail over across providers, reducing blast radius from any single outage.



Regulatory Flexibility

Easier to keep data in-region and meet sovereign/compliance requirements market by market.

DISADVANTAGES



Operational Complexity

Different consoles, APIs, and services per provider multiply what teams must learn and maintain.



Networking & Cost Overhead

Cross-cloud data transfer adds latency and egress fees that a single-cloud setup avoids.



Inconsistent Security & IAM

Each provider has its own identity, policy, and monitoring model — consistency takes deliberate effort.



Higher Tooling & Skills Cost

Multi-cloud management platforms and cross-platform expertise add real, ongoing spend.

innovate

achieve

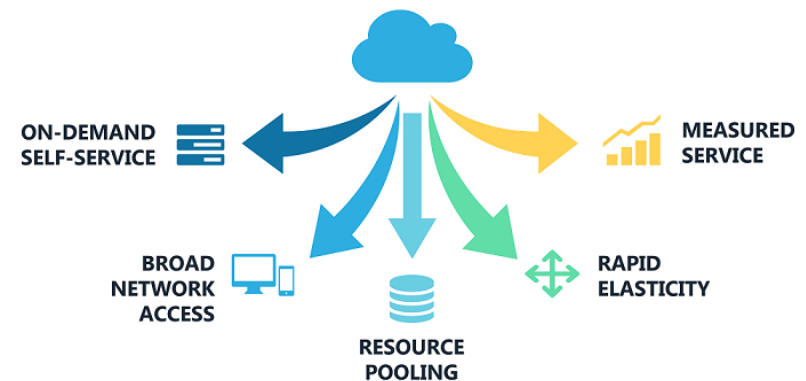
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Cloud Essential Characteristics



Essential Characteristics

5 Essential Characteristics of Cloud Computing

Ref: The NIST Definition of Cloud Computing

<http://csrc.nist.gov/publications/nistpubs/800-145/SP800-145.pdf>



On-demand Self-service

A consumer can unilaterally provision computing capabilities, such as server time and storage, as needed automatically without requiring human interaction with each service provider.



Ubiquitous Network Access

Capabilities are available over the network and accessed through standard mechanisms that promote use by heterogeneous thin or thick client platforms (e.g., mobile phones, tablets, laptops, and workstations).



Location Transparent Resource Pooling

The provider's computing resources are pooled to serve multiple consumers using a multi-tenant model, with different physical and virtual resources dynamically assigned and reassigned according to consumer demand.



Rapid Elasticity

Capabilities can be elastically provisioned and released, in some cases automatically, to scale rapidly outward and inward commensurate with demand. To the consumer, the available capabilities appear to be unlimited.



Measured Service with Pay per Use

Cloud systems automatically control and optimize resource use by leveraging a metering capability at some level of abstraction appropriate to the type of service (e.g., storage, processing, bandwidth, and active user accounts). Resource usage can be monitored, controlled, and reported, providing transparency for both the provider and consumer.



Cloud computing delivers on-demand access to a shared pool of configurable resources with speed, scale, and efficiency.

Resource pooling

Feature	Description
 Shared Resource Pool	Compute, storage, networking, and memory are dynamically allocated from a common pool of resources.
 Multi-Tenancy	Multiple customers securely share the same physical infrastructure while remaining logically isolated.
 Economies of Scale	Shared infrastructure significantly reduces operational costs for both providers and customers.
 High Resource Utilization	Resources are continuously optimized to maximize efficiency and reduce idle capacity.
 Dynamic Allocation	Resources are automatically provisioned and released based on workload demand.
 Security & Privacy Isolation	Strong isolation mechanisms ensure customer data and workloads remain secure despite shared infrastructure.
 Global Availability	Resources can be provisioned from geographically distributed data centers for improved performance and resilience.

Cloud providers maintain large pools of computing, storage, and networking resources that are shared among multiple customers using a secure multi-tenant architecture. Resources are dynamically allocated and released based on demand, enabling high utilization, cost efficiency, scalability, and reliable service delivery while ensuring strong security and workload isolation.

Broad Network Access

Feature	Description
 Anywhere Access	Access applications securely from any location with Internet connectivity.
 Device Independence	Services work across desktops, laptops, tablets, smartphones, and IoT devices.
 Standard Network Protocols	Services are accessed through web browsers, mobile applications, or APIs using standard Internet protocols (HTTPS, REST, gRPC).
 Near Real-Time Access	Applications respond within seconds or milliseconds depending on workload and network conditions.
 Secure Connectivity	Authentication, encryption, VPNs, and Zero Trust security protect user access.
 Implementation Transparency	Users interact with cloud services without needing to know where or how the infrastructure is implemented.
 Self-Service Access	Users can provision and access services through web portals, mobile apps, APIs, or command-line interfaces.

Ubiquitous Network Access enables cloud services to be securely accessed from anywhere, at any time, using standard Internet technologies and a wide variety of connected devices. This device-independent access allows users to work, collaborate, and consume cloud services seamlessly across locations and platforms while remaining unaware of the underlying infrastructure.

On Demand Self Service

Feature	Description
 Instant Provisioning	Resources can be created or released within seconds or minutes.
 Self-Service Portal	Users provision resources through web consoles without provider intervention.
 API-Driven Automation	Cloud services expose APIs for automated provisioning and management.
 Infrastructure as Code (IaC)	Resources can be deployed using tools such as Terraform, Bicep, CloudFormation, or Pulumi.
 Automated Provisioning	Policies, workflows, and AI-driven automation enable hands-free deployment.
 Accessible Anywhere	Services can be provisioned securely from any location with Internet connectivity.
 Standard Protocols	Cloud platforms expose RESTful APIs, HTTPS, SDKs, and CLI tools for integration.

On-Demand Self-Service enables users to provision, configure, and manage cloud resources whenever required through self-service portals, APIs, or automated Infrastructure as Code tools. This eliminates manual provisioning, accelerates application delivery, and provides organizations with the agility to respond quickly to changing business demands.

Rapid Elasticity

Feature	Description
 Instant Scaling	Resources can be added or removed within seconds or minutes.
 Scale Out / Scale In	Increase or decrease the number of resources automatically based on demand.
 Automated Scaling	Auto Scaling policies monitor workloads and adjust capacity without manual intervention.
 Dynamic Resource Allocation	Compute, storage, and networking resources are continuously reassigned based on workload requirements.
 Cost Optimization	Organizations pay only for the resources they actually consume.
 Faster Deployment	New infrastructure and application environments can be provisioned rapidly.
 Global Scalability	Applications can expand seamlessly across multiple regions and availability zones.

Rapid Elasticity is the capability of a cloud platform to automatically provision and release computing resources based on workload demand. Resources can be scaled up or down within seconds, allowing applications to handle unpredictable traffic while optimizing performance, availability, and cost. To users, cloud resources often appear virtually unlimited, even though they are dynamically allocated from a shared pool.

Metered by Use

Feature	Description
 Pay-as-You-Use Pricing	Customers pay only for the resources they consume.
 Usage Metering	Resource usage is continuously measured and recorded automatically.
 Cost Transparency	Detailed dashboards provide visibility into usage, billing, and spending trends.
 No Upfront Investment	No need to purchase servers or infrastructure before use.
 Stop Paying When Not in Use	Resources can be stopped or deleted to avoid unnecessary charges.
 Automated Billing	Billing is generated automatically based on measured consumption.
 Cost Optimization	Organizations can monitor utilization and eliminate unused or underutilized resources.

Measured Service is the capability of a cloud platform to automatically monitor, meter, and bill resource consumption. Customers pay only for the services they use, while detailed usage reports, billing dashboards, and cost management tools provide complete visibility and control over cloud spending.



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Summary – Wrap Up

Session
WRAP UP

Cloud Advantages

Key Benefits of Cloud Computing

Modern. Scalable. Cost-Effective.



Cloud vs Hosted

Cloud applications and web applications are related, but they are not the same.

A web application is an application that is accessed through a web browser.

A cloud application is designed to leverage cloud computing capabilities such as elastic scalability, high availability, automation, distributed architecture, and managed cloud services.

While many cloud applications provide a web interface, they may also expose mobile applications, APIs, serverless functions, and microservices.

All cloud applications may provide web-based access, but not all web applications are cloud-native applications.

Traditional Web Application

Hosted on one or a few servers

Manual scaling

Monolithic architecture

Manual deployments

Limited fault tolerance

Basic virtualization

Limited API integration

Cloud-Native Application

Runs across distributed cloud infrastructure

Automatic elastic scaling

Microservices architecture

CI/CD automated deployments

Self-healing and highly available

Containers, Kubernetes, and serverless

API-first architecture

Cloud Challenges

Cloud Provider Responsibilities

-  Physical data center security
-  Network infrastructure security
-  Hardware and virtualization security
-  High availability & disaster recovery
-  Compliance certifications
-  Threat detection and infrastructure monitoring

Cloud security is the practice of **protecting** cloud-based applications, data, infrastructure, and services through a combination of technologies, policies, and operational controls.

It is based on a **Shared Responsibility Model**, where cloud providers secure the infrastructure while customers are responsible for protecting their identities, applications, data, and configurations.

Modern cloud security emphasizes Zero Trust, encryption, identity management, continuous monitoring, and compliance to ensure confidentiality, integrity, and availability of cloud resources.

Customer Responsibilities

-  Identity & Access Management (IAM)
-  Strong passwords & Multi-Factor Authentication (MFA)
-  Data encryption and key management
-  Secure application configuration
-  Data governance and regulatory compliance
-  Patch applications and manage endpoints

Common Cloud Security Threats

-  Phishing and credential theft
-  Weak or reused passwords
-  Misconfigured cloud storage or services
-  Unauthorized access due to excessive permissions
-  Malware and ransomware attacks
-  Distributed Denial of Service (DDoS) attacks
-  Data breaches and accidental data exposure
-  Insider threats

Cloud Failures

Major Public Cloud Outages (Past 5 Years)

Key outages that impacted millions of users and businesses worldwide



DATE	PROVIDER	SCOPE / REGION	DURATION	IMPACTED SERVICES	KEY IMPACT
Dec 7, 2024	aws	US-EAST-1	Several hours	EC2, S3, RDS, CloudFront, Lambda, DynamoDB	Disney+, Netflix, Slack, Robinhood, Coinbase, and more services affected
Jun 13, 2024		Global	~2 hours	Cloud Run, Cloud SQL, GKE, BigQuery, Firebase	Widespread impact on apps and enterprise workloads
Feb 28, 2023		Global	~3+ hours	Multiple services including Virtual Machines, Storage, Azure AD	Enterprise applications and user sign-ins impacted
Dec 13, 2022	aws	US-EAST-1	~4 hours	EC2, Lambda, RDS, DynamoDB, SQS	High error rates and timeouts across many applications
Oct 20, 2021		Global	~2 hours	Compute Engine, GKE, BigQuery, Cloud Storage	Major disruption to workloads across regions
Jul 18, 2021		Global	~1+ hour	Azure AD, Storage, SQL Database, Key Vault	Sign-in issues and service degradation for customers

BY THE NUMBERS

 **Millions**
of users affected across the globe

 **Hours**
of downtime and service disruption

 **Global Impact**
Across industries, applications and critical services

 **Resilience Matters**
Redundancy, multi-region architecture and robust monitoring are key



Cloud outages are inevitable, but resilience is in our control.

Organizations must design for high availability, monitor proactively and have a strong incident response plan.

Source: Status pages, Public reports, Media coverage



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Points to Ponder

Session
WRAP UP

Points to Ponder

1. Do you think Cloud is a boon or a bane
2. What is your trust level of any cloud you use
3. What do you Understand by
 1. Hosted Application
 2. Cloud enabled
 3. Cloud native
 4. Cloud Agnostic
4. What do you think will be your main considerations when choosing a cloud provider?